

Maximum Entropy Spectral Estimation

Script: Chapter 11 (AR models, Yule–Walker, Levinson–Durbin, Burg)

Learning goals

- Describe a resonance by an autoregressive (AR) model and relate the AR poles to resonance frequency and damping.
- Derive and solve the Yule–Walker equations with the Levinson–Durbin recursion; implement Burg’s method; explain the maximum entropy principle.
- Compare MEM (AR) spectra with windowed periodograms on the very short records available in a knock window (32–128 samples): frequency accuracy, peak shape, resolution, artefacts.
- Select the model order (FPE, AIC, MDL) and recognise the symptoms of a wrong order (spurious peaks, line splitting).

Background

A knock window is short: 60° crank angle are 3.3 ms at 3000 rpm and only 1.7 ms at 6000 rpm, i.e. 42–83 samples at $f_s = 25$ kHz (the rate of Sheet 10). The DFT of N samples has the bin spacing f_s/N (781 Hz for $N = 32$), and windowing widens the peaks further. Parametric methods fit a model to the data and are not bound to this “Rayleigh limit”.

An AR(p) process is white noise $e[n]$ (variance σ^2) filtered by an all-pole filter:

$$x[n] = -\sum_{i=1}^p a_i x[n-i] + e[n], \quad A(z) = 1 + \sum_{i=1}^p a_i z^{-i}, \quad P(f) = \frac{\sigma^2 T_s}{|A(e^{j2\pi f/f_s})|^2}.$$

Multiplying by $x[n-m]$ and taking expectations gives the *Yule–Walker equations* $\sum_{i=1}^p a_i r[m-i] = -r[m]$, $m = 1, \dots, p$, and $\sigma^2 = r[0] + \sum_{i=1}^p a_i r[i]$. The *Levinson–Durbin recursion* solves them order by order in $O(p^2)$ operations ($E_0 = r[0]$):

$$k_m = -\frac{r[m] + \sum_{i=1}^{m-1} a_i^{(m-1)} r[m-i]}{E_{m-1}}, \quad a_i^{(m)} = a_i^{(m-1)} + k_m a_{m-i}^{(m-1)}, \quad a_m^{(m)} = k_m, \quad E_m = (1 - k_m^2) E_{m-1}.$$

Burg’s method uses the same order recursion, but estimates every reflection coefficient k_m directly from the data by minimising the sum of forward and backward prediction error powers,

$$k_m = -\frac{2 \sum_{n=m}^{N-1} f_{m-1}[n] b_{m-1}[n-1]}{\sum_{n=m}^{N-1} (f_{m-1}^2[n] + b_{m-1}^2[n-1])}, \quad \begin{aligned} f_m[n] &= f_{m-1}[n] + k_m b_{m-1}[n-1], \\ b_m[n] &= b_{m-1}[n-1] + k_m f_{m-1}[n], \end{aligned}$$

with $f_0[n] = b_0[n] = x[n]$. No autocorrelation estimate (and hence no implicit windowing of the data) is needed. Order selection criteria (E_p = prediction error power of order p , N = record length):

$$\text{FPE}(p) = E_p \frac{N+p+1}{N-p-1}, \quad \text{AIC}(p) = N \ln E_p + 2p, \quad \text{MDL}(p) = N \ln E_p + p \ln N.$$

Code: `code/jslinux/sheet11/`: `ar.h` (AR library), `fft.h` (radix-2 FFT with the API of Sheet 8, `fft(re, im, n, inverse)` – you may use your own), `memspec.c` (Exercise 11.2), `resolve.c` (Exercise 11.3).

Exercise 11.1 AR models and the maximum entropy principle

PAPER

- AR(1): derive a_1 and σ^2 from $r[0], r[1]$ and the PSD. For which sign of a_1 is the process low-pass, for which high-pass?
- AR(2): write down the Yule–Walker equations and solve them for a_1, a_2 in terms of $\rho_1 = r[1]/r[0], \rho_2 = r[2]/r[0]$. Then run the Levinson–Durbin recursion by hand for the normalised autocorrelation $r = [1, -0.0627, -0.8973]$ and give $k_1, k_2, a_1, a_2, E_1, E_2$.
- Show that an AR(2) model with complex poles $z_{1,2} = re^{\pm j\theta}$ has $a_1 = -2r \cos \theta, a_2 = r^2$ and describes a resonance. Compute r, θ, a_1, a_2 for the knock mode ($f_1 = 6.5$ kHz, decay time $\tau = 0.8$ ms, $f_s = 25$ kHz) and compare with b). Estimate the -3 dB bandwidth of the AR(2) peak from r .
- Maximum entropy.* Given $r[0], \dots, r[p]$, there are infinitely many PSDs consistent with these values. Burg proposed to choose the one with maximum entropy rate $h = \frac{1}{4\pi} \int_{-\pi}^{\pi} \ln P(\omega) d\omega$ (+const.). Show with a Lagrange argument that this PSD has the form $P(\omega) = 1 / \sum_{|m| \leq p} \lambda_m e^{-j\omega m}$, i.e. it is an AR(p) spectrum. How does the MEM spectrum treat the unknown lags $|m| > p$, compared with a (windowed) periodogram?
- Show that Burg’s reflection coefficients always satisfy $|k_m| \leq 1$. Why does this guarantee a stable (minimum-phase) $A(z)$ and why is that important? Does the Yule–Walker method with the *biased* autocorrelation estimate have the same property?

Exercise 11.2 MEM vs. periodogram on short knock windows

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```
1 gcc -O2 -o memspec memspec.c -lm && ./memspec [N] [p] # default N = 64, p = 8
```

- Complete `ar_levinson()`, `ar_burg()` and `ar_psd()` in `ar.h`. Test them with the AR(2) example of 11.1 b) (you must get $a_1 = 0.1194, a_2 = 0.9048$).
- Part (a) of `memspec` takes one knock window (N samples starting 10° after TDC, 3000 rpm, mean removed) and computes the Hann periodogram (zero padded to 1024), the Yule–Walker and the Burg spectrum. Plot `mem.csv` for $N = 32, 64, 128$ and describe the differences.
- Part (b) estimates the knock frequency (maximum between 5.5 and 7.5 kHz) in 300 windows for $N = 32, 64, 128$ with four methods and reports mean, standard deviation, rms error, the -3 dB width of the peak and how often the second mode (10.5 kHz) is found. Interpret the table: in which respect is MEM better than the periodogram, in which not? Why does Burg’s frequency estimate scatter more than the periodogram’s?
- Complete `ar_crit()`. Part (c) applies FPE, AIC and MDL to Burg models up to order 20. Which orders are chosen, and why are about 6–8 poles plausible for the knock signal?
- Part (d) uses Burg with orders 2 to 32 for $N = 64$. Explain the growing number of peaks and the “split” knock peak, and give a rule of thumb for the order.

Exercise 11.3 Resolving two close resonances

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A temperature-dependent shift of the knock mode in one cylinder, or a second structural mode of the engine block, can create two resonances only a few hundred Hz apart. Can the ECU still separate them within one knock window?

```
1 gcc -O2 -o resolve resolve.c -lm && ./resolve 500 && ./resolve 250 && ./resolve 1000
```

- Complete `resolved()`: the two components count as resolved if there is a local maximum within $\Delta f/2$ of f_1 and one within $\Delta f/2$ of f_2 , and the spectrum dips by at least 3 dB between them.

- b) The program generates two sinusoids at $f_1 = 6.5 \text{ kHz}$ and $f_1 + \Delta f$ (random phases) in white noise with an SNR of 40, 20 and 10 dB per component and records how often the rectangular periodogram, the Hann periodogram and Burg ($p = N/3$, at most 32) resolve them for $N = 16 \dots 256$. Determine the minimum record length for $\geq 90\%$ resolution for $\Delta f = 500 \text{ Hz}$ and compare with the Rayleigh limit $N \approx f_s / \Delta f$. How does the SNR influence the result?
- c) Repeat for $\Delta f = 250$ and 1000 Hz . At $\Delta f = 500 \text{ Hz}$ and 6000 rpm , a 60° window has 42 samples: which method can resolve the two modes?
- d) Part (b) of the program uses *damped* modes with $\tau = 0.8 \text{ ms}$, as in a real knock burst. Explain why now even long records and high SNR do not help for $\Delta f = 250 \text{ Hz}$ and why Burg's success rate saturates at about 50–60% for $\Delta f = 500 \text{ Hz}$.

Deliverables

11.1 on paper. Completed `ar.h`; plots of `mem.csv` for $N = 32, 64, 128$; the tables of `memspec` parts (b)–(d) and of `resolve` for $\Delta f = 250, 500, 1000 \text{ Hz}$ with a half-page interpretation: when would you use MEM in a knock controller, and when not?